BIG DATA, INDIVIDUAL ASSIGNMENT 1

Matrix Multiplication

Basic implementations in Python, Java and C

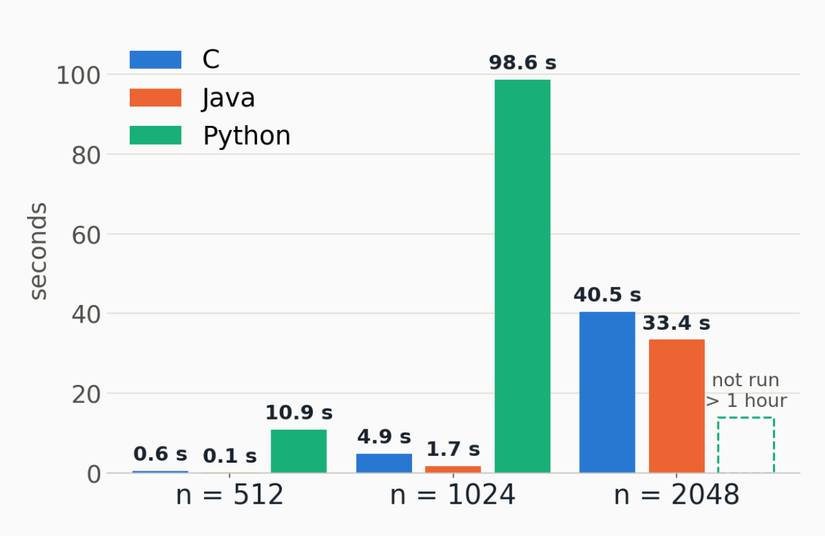
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Grado en Ciencia e Ingeniería de Datos, 2026-2027

How does the time grow with bigger matrices?

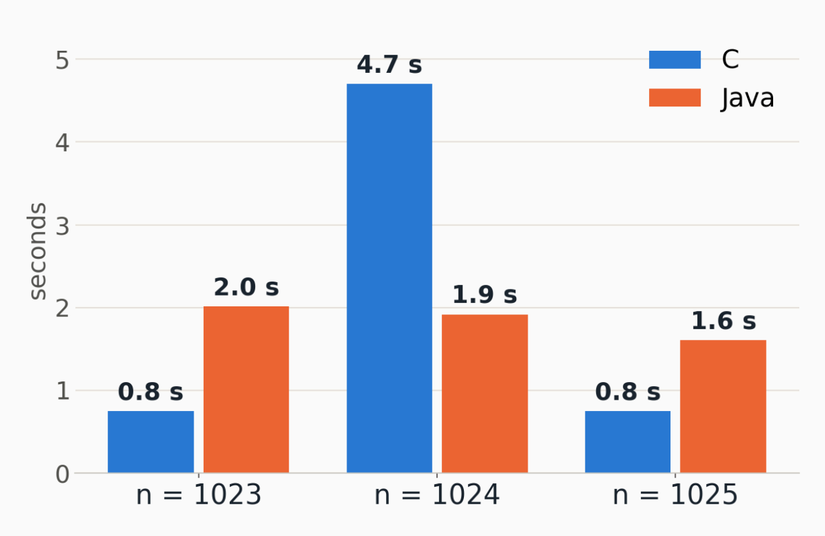
Same algorithm (3 loops) in Python, Java and C. Fair test: same numbers, same computer, 5 runs for each size.

Time when n gets bigger



2 times bigger, about 8 times more time. Python is the slowest.

The surprise: C at n = 1024

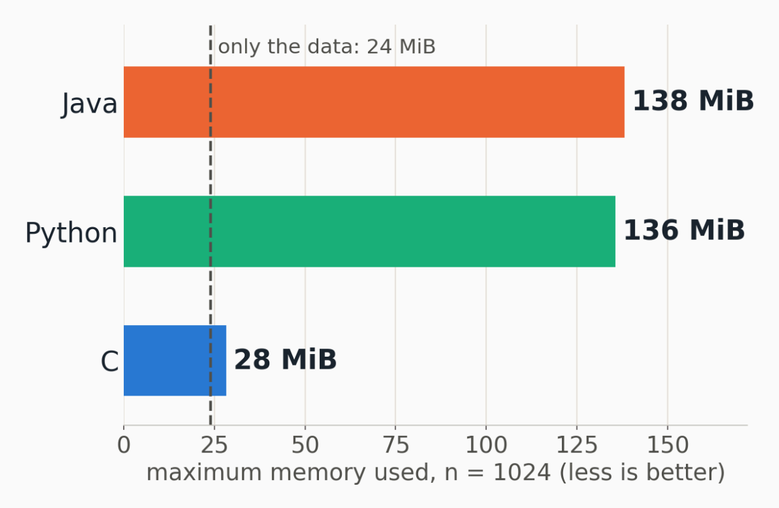


Only 1 number of difference, but C is 6 times slower. The reason is the cache.

Conclusions

× 8	2 times bigger matrix, about 8 times more time.
Python	The slowest: 18 to 190 times slower than C.
C vs Java	C is faster, but not with powers of two (cache).
Memory	C uses the least. Java and Python need more.

Memory for a 1024 × 1024 matrix



C uses almost only the data. Java and Python use almost 6 times more.

Code, data and report: github.com/Pablo5250/matrix-multiplication